University of South Carolina Interview Helper Application

Software Requirements Specification

CSCE 247: Software Engineering

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Introduction

Purpose

In recent years it seems newer Computer Science and Computer Information Systems students are failing technical interviews, be it for internships or jobs. Many are finding that the most important thing to do is practice, citing that one of the most common things that cause failure in technical interviews is being underprepared and overconfident. Data from a 2025 study of FAANG candidates showed that students with less than 5 technical interviews had only a 40% chance of passing interviews, while those with more practice had an 81% success rate. Raw technical skill in isolation showed little effect on interview outcomes, highlighting that practice in communicating solutions under pressure is essential.

https://interviewing.io/blog/how-know-ready-interview-faang

Stakeholders

- University of South Carolina - Students coming from UofSC reflect on the university as a whole
- University of South Carolina students - The students themselves are the ones taking the interviews
    - Computer Science majors
    - Computer Information Systems majors
- Companies and Businesses looking to hire - Companies are having to turn down students because they lack the skill necessary to hire

Personas

Image credit: iStock

Darryl Holiday

Age: 18
Year: Freshman
Major: Computer science
Type: Teacher

Goals: Wants to have an internship where he gets hands on experience, also interested in teaching.

Bio: Military family background, first generation college student, likes to take walks around the city.

Motivation: Wants to succeed because he is the first in his family to have college education, knows that he needs an internship to get ahead in a competitive field.

Needs: Wants to know specifics about competition for internships and what successful people have done to get where they are.

Image credit: iStock

Alyssa Middleton

Age: 19
Year: Junior
Major: CIS
Type: Leader

Goals: Wants to participate in research regarding networking and cybersecurity.

Bio: Graduated high school a year early at the top of her class, works part time and commutes to campus from Cayce.

Motivation: Wants to research so that she can make connections with professors that could give her a headstart in the industry.

Needs: She needs to know which interviews and research opportunities are most relevant to her skills.

Image credit: vecteezy

George Booker

Age: 28
Year: Senior
Major: Computer science
Type: Listener

Goals: Wants to find a job involving programming, wants to become more charismatic.

Bio: Took a few years to work after high school, lives in East Columbia with his wife, has loved working with computers since childhood.

Motivation: Wants to be able to support himself and his family, his last job damaged his passion for computers and he wants to gain that back

Needs: A platform that will help him market himself and be more charismatic because he is socially awkward and nervous.

Constraints

Time Constraints

- This project must be completed by late April, 2026.

Monetary Constraints

- This project has a budget of $0.

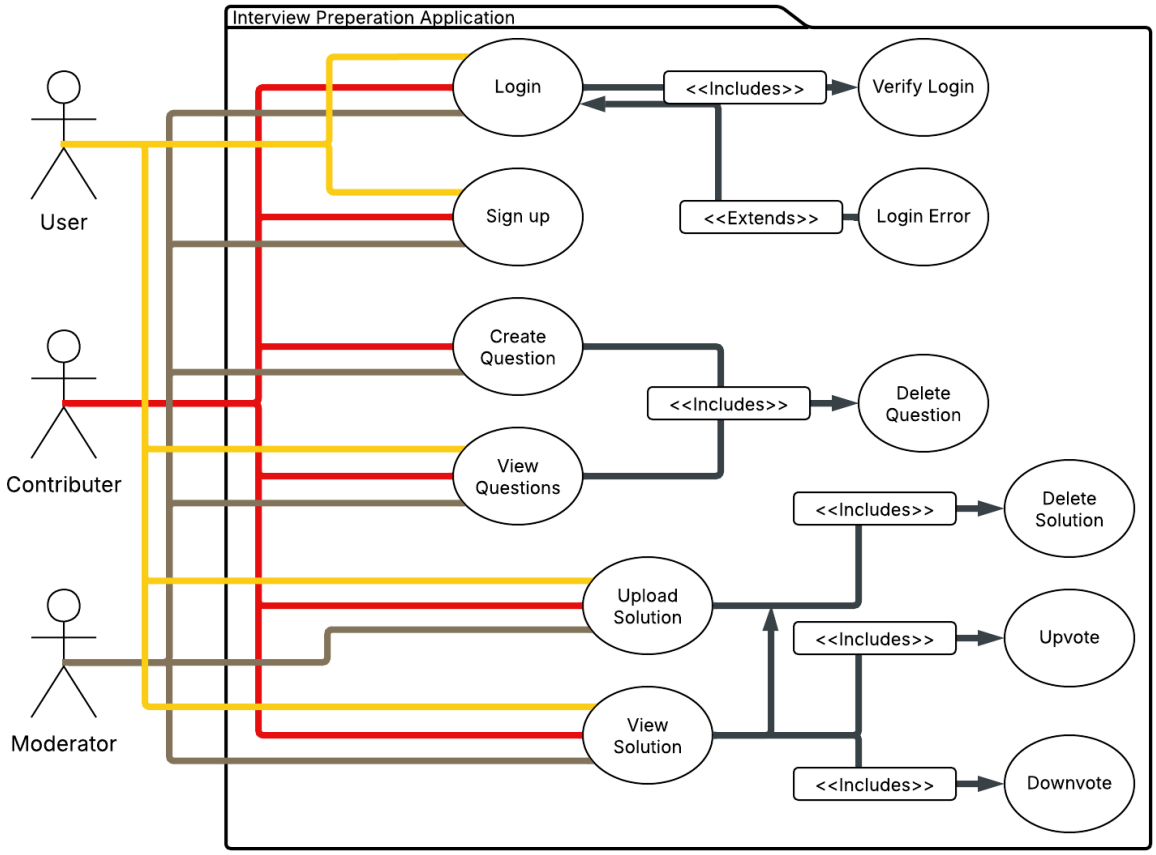
Technical Constraints

- The code must be done in Java.
- The project must be accessible on desktop and mobile platforms.

Overall Description

The program must be accessible to users in the USC system. Users will be able to access the program from desktop or mobile platforms. The program will contain data from technical interviews. The user will be given a series of technical interview questions, with intermittent standard interview questions. The program will contain data from the user's previous answers, tracking their progress and giving them a readiness score.

Business Use Cases



Functional Requirements

Functional requirements spreadsheet

Non-Functional Requirements

Look and Feel Requirements

- The application shall have a user interface that is easy for beginners to use.
- The application shall use USC's color palette

Usability Requirements

- The application shall be usable by students who have completed one computer science class
- The system shall give clear error messages (i.e. "Password must contain at least one number")

Performance Requirements

- The application shall support one thousand concurrent users
- The application shall be able to run on desktop

Maintenance Requirements

- The application shall be modular to allow future additions
- The application's code shall be well documented and commented

Cultural Requirements

- The application shall use inclusive language
- The application shall use an interface that supports users of diverse backgrounds

Competitive Analysis

	CodeSignal
Strengths	CodeSignal provides users with a variety of course style materials to teach and reinforce programming and design strategies. The service also provides an analysis of one's ability through assessments. CodeSignal also provides a platform for interviewers to conduct their interviews, ensuring that the interviewee is already familiar with the environment. CodeSignal's close relationship with interviewers produces more relevant assistance to customers.
Weaknesses	Some users complain about the user interface. The most common complaints are about unclear instructions and lacking organization. Users also mention a lack of extensive free options in their reviews.
Implications (For us)	Curating information specific to the companies interviewees wish to prepare for is the most potent strength of this service.

	Pramp
Strengths	Pramp prepares interviewees for technical interviews through real-time peer interviews, allowing users to share their own experiences and information, improving their overall learning. Because of its basis in peer review and interviews, more of Pramp's features are able to be accessed freely while remaining up to date with interview practices.
Weaknesses	Users note a lack of a built-in development environment, introducing a friction in the process of starting. Users also note a wide variation in results due to the variation in peer quality.
Implications (For us)	Providing a platform to allow for users to assist each other increases the flexibility of the service while reducing costs.

	InterviewBit
Strengths	InterviewBit provides a variety of structured lessons to ensure users have practice in many different components of software production. Users are provided with training specific to companies of interest, including mock interviews. InterviewBit provides users with the opportunity to interact with experienced engineers and receive feedback.
Weaknesses	Users note a lack of variety in question type and difficulty when compared to

	alternatives. Difficulty using and navigating the site as well as slowness in the video and interview components is also mentioned.
Implications (For us)	Allowing users to practice with experienced developers and in information relevant to a specific company increases the value of the service.

Summary

	Strengths	Weaknesses	Implications
CodeSignal	Mock interviews Company specific information	Cost User interface	Collect information on specific companies
Pramp	Peer based training Flexibility	Peer quality Lack of development environment	Provide collective training
InterviewBit	Mock interviews Experienced developers	Lack of variety Difficult user interface Inconsistent video	Experienced developers.

Based on all the collective strengths and weaknesses found in what could be competitors to our application. There should be an importance placed on an easy to use and understand interface with a wide variety of questions from those with experience allowing for a verisimilitude of both practical development training and practical interview training. As seen in all three competitors, user interface or parts of the user interface and or experience are some of the most common issues, so a big focus of our application should be how can we make certain parts more intuitive and easier to manage for both new and returning users. As for strengths we noticed not only placing an importance on practical coding training, but on mock interviews from those with experience is highly valued, be it from companies or experienced developers. This means our application should also be able to provide solid mock interviews from those experienced, like so many of the teachers at UofSC, and a wide range of interview questions that focus on all aspects and possibilities. Overall there are three important things that should be present in our application based on what was analyzed from our prospective competitors: Strong mock interviews, an easy to use and understand interface, and finally not just focusing on one aspect of the process but both practical and social.